

Chapter 9

The Kerala School of Mathematics

The Nila Renaissance: Discovery of Calculus in Medieval India
(14th – 16th Century)

Introduction: The Nila Mathematics

- **Definition:** A remarkable school of mathematics and astronomy founded in the 14th century, which thrived until the 16th century in the southern tip of India.
- **Significance:** Long before Newton and Leibniz formalized calculus in Europe, the Kerala School made pioneering discoveries in infinite series, trigonometric functions, and the foundations of what we now recognize as mathematical analysis.
- **The "Nila" Connection:** Often referred to as "Nila Mathematics," this intellectual movement was deeply rooted in a specific geographical landscape that provided the stability and isolation necessary for sustained intellectual inquiry.
- **Notable Mathematicians:** Madhava, Narayana, Nilakantha Somayaji, etc.

The Geographical Setting: The Nila River

Riverside Scholar-Hubs

The Kerala School flourished in a specific geography along the banks of the **Bharathappuzha (Nila) River** in central Kerala.



- **The Bharathappuzha (Nila) River:** The prominent mathematicians of this school predominantly lived and worked in a small geographical radius along the banks of the Bharathappuzha River (often affectionately called the Nila), specifically in the present-day **Malappuram and Palakkad** districts.
- **Tirunavaya Temple:** The region around Tirunavaya, situated on the riverbanks, was a renowned center of traditional Vedic and astronomical learning. It served as a cultural hub that fostered a deep, rigorous academic environment.
- **Geographical Isolation as a Shield:** During the 14th to 16th centuries, much of Northern India was experiencing severe political upheaval and invasions, which disrupted traditional centers of learning. **Kerala's geographical isolation, protected by the Western Ghats, provided a peaceful haven.** This allowed for an unbroken *Guru-Shishya parampara* (teacher-student tradition) lasting centuries.

Madhava of Sangamagrama

The Founder (c. 1340 – 1425)

- The Founder: Widely regarded as the founder of the Kerala School. "Sangamagrama" translates to the modern-day town of Irinjalakuda.
- Key Contributions: He is credited with the "decisive step" from the finite procedures of ancient algebra to the infinite limit-passages of analysis.
- Considered as the Father of Indian Calculus.
- His work on power series expansions predates European discoveries by over two centuries.
- Calculated the value of π to 11 decimal places using his power series.
- Notable Formula: The Madhava-Leibniz series for π



Infinite Series for π (Pi)

The Madhava-Leibniz Series

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

This series calculates π with any desired precision, a monumental leap in analytical mathematics.

Precision & Correction

Madhava calculated π to 11 decimal places (3.14159265359) using only 13 terms of his series plus a correction term.

Correction Term Formula:

$$F_n = \frac{n^2 + 1}{4n^3 + 5n}$$

Narayana Pandit: The Bridge

Combinatorics & Sequences

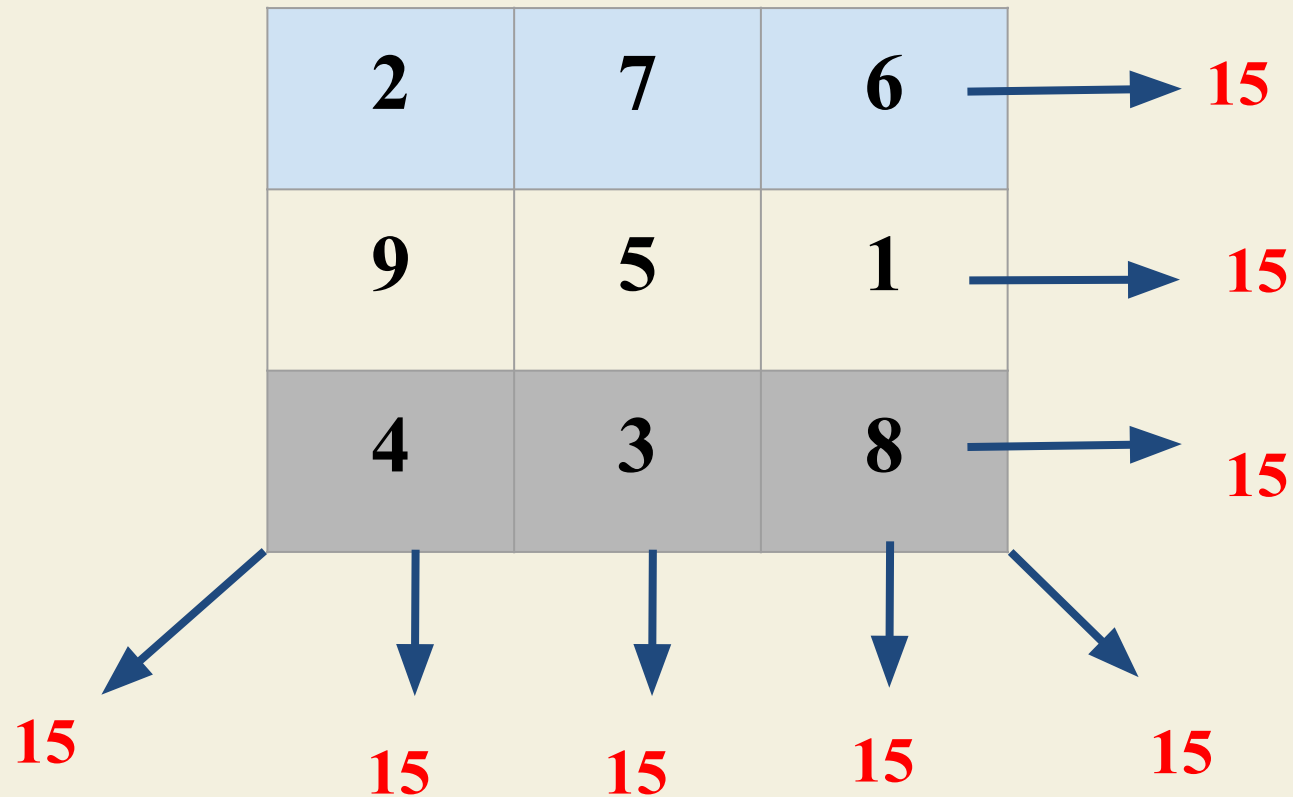
Author of the **Ganita Kaumudi** (1356), Narayana Pandit bridged classical Indian math with the Nila Renaissance.

- **Permutations:** Developed rules for the systematic generation of all permutations.
- **Magic Squares:** Investigated rules for perfect magic squares and circles.
- **Sequences:** Proposed the "Narayana's Cows" sequence (similar to Fibonacci).



Magic Square: Proposed by Narayana Pandit

- Definition: A magic square is a square grid filled with distinct numbers (usually positive integers) where the sum of the numbers in each row, each column, and both main diagonals is the same constant.
- This constant is often referred to as the "magic constant" or "magic sum".
- Example:



Magic Square Formula

A formula for creating matrix square of order 4 for various values of n.

$n-3$	1	$n-6$	8
$n-7$	9	$n-4$	2
6	$n-8$	3	$n-1$
4	$n-2$	7	$n-9$



More Examples

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10	3	13	8
5	16	2	11
4	9	7	14
15	6	12	1

12	3	6	13
14	5	4	11
7	16	9	2
1	10	15	8

Nilakantha Somayaji

Authored The Tantrasamgraha (1501)

- Born in Tirur, heavily associated with the Nila region. A prolific writer and observational astronomer.
- **Key Contributions:** Author of the monumental **Tantrasamgraha (1501)**, a major treatise on astronomy and mathematics.
- Developed a geo-heliocentric model of the solar system where the planets orbit the Sun, which in turn orbits the Earth. This model mathematically anticipated the one proposed by Tycho Brahe in Europe almost a century later.
- Expanded upon Madhava's infinite series and provided the philosophical and mathematical justifications for them.

The Yuktibhasha: Authored by Jyeshthadeva

Discourse on Rationale (c. 1530)

- Authored by **Jyeshthadeva**, this text is considered the world's first calculus textbook.
- A student of Damodara (who was in turn a student of Nilakantha).
- Unlike previous cryptic verses, it provides detailed prose explanations and rigorous proofs (*Yukti*).
- **Language:** Unlike previous texts written in esoteric Sanskrit verse, Jyeshthadeva wrote in Malayalam, the regional language. This made the knowledge accessible beyond a small, elite circle.
- **Focus on Proof:** It is widely considered the world's first calculus text. While earlier Indian mathematicians often provided rules without proofs, the Yuktibhasha is dedicated entirely to *Yukti* (rationale or proof). It provides detailed, step-by-step derivations for the theorems of Madhava and Nilakantha.

Democratizing Math: Proof & Language



Malayalarajya

Written in **Malayalam** instead of Sanskrit. This made advanced math accessible to regional students and professional practitioners.



Systemic Proofs

Introduced the concept of *Upapatti* (demonstrative proof). It moved from "what is the rule" to "why does the rule work."



Calculus Origins

Contains the derivation of the power series for inverse tangent, now known as the Gregory-Leibniz series.

Legacy: The Birth of Calculus

- ✓ **Foundations of Analysis:** The Kerala School developed the core ideas of calculus—infinite series and integration—200 years before Europe.
- ✓ **Global Impact:** Current research explores the possibility of knowledge transfer via Jesuit missionaries from Malabar to Europe.
- ✓ **Scientific Method:** The shift from purely religious astronomy to mathematical "Rationale" (*Yukti*) marked a pre-modern scientific revolution.
- ✓ **The Nila Heritage:** The banks of the Nila River remain a testament to India's sophisticated mathematical heritage.



Questions?

Thank you for your attention.